

ROTHERA STATION, Antarctica

WMO: 890620

Lat: 67.566S

Lon: 68.130W

Elev: 33

StdP: 100.93

Time Zone: -3.00 (W03)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	-23.8	-21.4	-27.3	0.3	-23.4	-24.7	0.4	-20.8	22.0	-4.5	20.2	-4.7	3.1	0	1.108

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.9	4.5	1.7	3.8	1.3	3.2	0.9	2.3	3.8	1.8	3.3	1.4	2.8	5.7	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
0.6	4.0	2.5	0.2	3.8	2.0	-0.1	3.8	1.7	13.9	4.1	13.0	3.5	12.3	2.9	5.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.2	17.1	15.1	-25.5	6.5	4.1	0.8	-28.5	7.0	-30.9	7.5	-33.2	8.0	-36.3	8.6		
(5)			-25.7	3.7	3.8	0.8	-28.4	4.3	-30.6	4.8	-32.7	5.3	-35.5	5.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.8	1.4	0.8	-0.7	-2.9	-3.8	-6.6	-8.7	-9.5	-7.7	-5.0	-2.5	0.1		
	DBStd	5.33	1.23	1.52	2.16	2.83	3.38	4.62	5.50	5.96	6.23	4.29	2.73	1.67		
	HDD10.0	5029	265	258	333	387	427	498	580	604	532	466	375	306		
	HDD18.3	8071	523	491	591	637	685	748	838	863	782	724	625	564		
	CDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	6.4	4.6	5.7	6.3	6.5	7.3	7.1	6.7	6.8	7.7	7.5	6.2	4.9		
(15) Precipitation	PrecAvg	678	56	59	70	59	61	45	43	50	59	68	60	48		
	PrecMax	904	98	78	110	93	104	97	77	96	80	130	101	107		
	PrecMin	510	31	31	36	24	27	18	18	11	36	27	21	11		
	PrecStd	108	18	14	18	17	22	18	15	19	14	26	25	21		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.6	4.9	4.6	2.9	3.4	1.9	1.8	1.3	2.0	2.4	3.3	4.8		
		MCWB	2.3	2.0	2.4	1.1	1.5	0.3	0.0	-0.1	0.4	0.6	0.7	1.1		
	2%	DB	4.6	4.0	3.4	1.9	2.0	1.0	0.6	0.4	1.1	1.5	2.0	3.7		
		MCWB	1.7	1.6	1.5	0.3	0.4	-0.4	-0.8	-0.7	-0.2	0.1	0.3	0.9		
	5%	DB	4.0	3.4	2.6	1.2	1.1	0.2	-0.2	-0.3	0.5	0.9	1.3	3.0		
		MCWB	1.3	1.2	0.9	-0.1	-0.1	-0.9	-1.3	-1.2	-0.6	-0.3	-0.3	0.5		
	10%	DB	3.3	2.8	2.0	0.6	0.5	-0.6	-1.2	-1.5	-0.2	0.3	0.7	2.5		
		MCWB	0.9	0.8	0.5	-0.6	-0.6	-1.6	-2.1	-2.4	-1.1	-0.7	-0.8	0.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.1	2.8	2.7	1.3	1.9	0.6	0.2	0.2	0.7	1.0	1.3	2.2		
		MCDB	4.9	4.2	4.3	2.4	3.3	1.6	1.3	0.9	1.5	1.9	2.6	3.8		
	2%	WB	2.2	2.1	1.7	0.7	0.7	-0.1	-0.5	-0.4	0.1	0.4	0.6	1.5		
		MCDB	3.9	3.5	3.1	1.6	1.6	0.8	0.4	0.3	0.8	1.3	1.7	3.1		
	5%	WB	1.8	1.6	1.2	0.2	0.1	-0.7	-1.2	-1.2	-0.4	0.0	0.1	1.0		
		MCDB	3.4	3.0	2.4	1.1	0.9	0.2	-0.3	-0.3	0.4	0.8	1.1	2.5		
	10%	WB	1.3	1.2	0.7	-0.4	-0.5	-1.6	-2.1	-2.4	-1.0	-0.6	-0.5	0.5		
		MCDB	2.9	2.5	1.8	0.6	0.4	-0.7	-1.2	-1.5	-0.2	0.3	0.6	2.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	2.9	3.0	2.7	2.8	3.3	4.1	5.5	5.6	5.3	4.1	3.4	3.0		
		MCDBR	3.7	3.6	3.1	3.4	3.6	4.0	5.7	5.1	4.7	3.9	3.8	3.7		
		MCWBR	2.6	2.5	2.4	3.0	3.0	3.7	5.3	4.7	4.4	3.5	3.1	2.6		
		MCDBR	3.4	3.3	3.0	3.2	3.5	3.8	5.9	5.0	4.8	3.7	3.6	3.4		
	5% WB	MCWBR	2.5	2.5	2.4	3.1	3.1	3.7	5.7	4.8	4.7	3.5	3.1	2.5		
(40) Clear-Sky Solar Irradiance	taub		0.278	0.273	0.255	0.217	N/A	N/A	0.149	0.208	0.245	0.262	0.268	0.277		
	taud		2.373	2.395	2.387	2.229	N/A	N/A	1.900	2.095	2.229	2.228	2.262	2.296		
	Ebn at Noon		976	922	832	620	N/A	0	163	595	816	920	979	991		
	Edh at Noon		103	88	67	39	N/A	0	8	43	75	102	114	116		
(44) All-Sky Solar Radiation	RadAvg		6.28	4.16	2.08	0.62	0.09	0.00	0.03	0.41	1.69	4.16	6.45	7.58		
	RadStd		0.30	0.28	0.14	0.05	0.01	0.00	0.00	0.03	0.10	0.16	0.30	0.46		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page